Ericsson PE Node WHI-PE-14 Routing Protocol Flap Resolution Standard Operating Procedure

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_3430

Description

This SOP outlines the critical steps to resolve the Routing Protocol Flap failure on Ericsson PE Node WHI-PE-14 at Whitefield East POP. The failure is characterized by a BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This phenomenon is attributed to the inherent complexity of routing protocols and the dynamic nature of network topology. In the Ericsson PE Node, the routing protocols (BGP, OSPF, and ISIS) operate in conjunction with the control plane, which is responsible for maintaining the routing table and forwarding information. When the routing protocols experience a flap, it can lead to an excessive exchange of routing updates, causing instability in the network. The root cause of this failure can be attributed to various factors, including misconfigured routing policies, hardware issues, or software bugs. It is essential to promptly identify and resolve the root cause to prevent network outages and ensure high network availability.

Pre-requisite Checks

- Verify that the remote hands have access to the Whitefield East POP and can perform physical checks on the node.

- Check the status of redundant links and ensure they are operational.

- Perform a thorough review of the site documentation, including circuit diagrams, to identify potential causes of the failure.

Potential Root Causes

Cause	Probability
Misconfigured routing policies	High
Hardware issues with the PE node	Medium
Software bugs in the routing protocols	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP router identifier 10.1.1.1, local AS number 100 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 192.168.1.2 4 100 0 0 0 0 0 00:00:00 0/0 , Description: This command displays a summary of the BGP neighbor table, including the number of messages received and sent, and the up/down status of each neighbor.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface ID Interface 192.168.1.2 1 FULL/BDR 00:00:32 Et0/0 GigabitEthernet0/0 , Description: This command displays a list of OSPF
```

```
neighbors, including their state, dead time, and the interface used to reach them.

Command: show isis neighbors, Expected Output: Neighbor State Pri Up Time Interface 192.168.1.2 UP 64 00:00:32 GigabitEthernet0/0 , Description: This command displays a list of ISIS neighbors, including their state, priority, up time, and the interface used to reach them.
```

Resolution Steps

Step 1: Step: Reset the BGP session, Command: clear ip bgp *, Description: This command resets the BGP session with all neighbors, which can help to clear out any misconfigured routing policies or software bugs.

Step 2: Step: Reconfigure the routing policies, Command: configure terminal router bgp 100 neighbor 192.168.1.2 remote-as 100 neighbor 192.168.1.2 route-map OUT in exit , Description: This command reconfigures the routing policies to ensure that they are correctly set up and will not cause any further routing flaps.

Step 3: Step: Check the PE node hardware, Command: show system resource, Description: This command displays a list of the PE node's hardware resources, including memory and CPU usage, which can help to identify any hardware issues.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP router identifier 10.1.1.1, local AS number 100 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 192.168.1.2 4 100 0 0 0 0 0 00:00:00 0/0 , Description: This command is used to verify that the BGP session has been successfully reset and that the routing policies are correctly configured.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface ID Interface 192.168.1.2 1 FULL/BDR 00:00:32 Et0/0 GigabitEthernet0/0 , Description: This command is used to verify that the OSPF neighbor table is correctly populated and that the routing policies are correctly configured.

Command: show isis neighbors, Expected Output: Neighbor State Pri Up Time Interface 192.168.1.2 UP 64 00:00:32 GigabitEthernet0/0 , Description: This command is used to verify that the ISIS neighbor table is correctly populated and that the routing policies are correctly configured.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy nvram:startup-config flash:, Description: This command restores the original configuration from the startup-config file in NVRAM to the running configuration.

- Step: Reapply the corrective measures, Command: configure terminal router bgp 100 neighbor 192.168.1.2 remote-as 100 neighbor 192.168.1.2 route-map OUT in exit , Description: This command reapplies the corrective measures that were previously performed to fix the issue.